

Dynamic mesh optimization for polygonized implicit surfaces with sharp features

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The paper presents a novel approach for accurate polygonization of implicit surfaces with sharp features. The approach is based on mesh evolution toward a given implicit surface with simultaneous control of the mesh vertex positions and mesh normals. Given an initial polygonization of an implicit surface, a mesh evolution process initialized by the polygonization is used. The evolving mesh converges to a limit mesh which delivers a high-quality approximation of the implicit surface. For analyzing how close the evolving mesh approaches the implicit surface, we use two error metrics: the metrics measure deviations of the mesh vertices from the implicit surface and deviations of mesh normals from the normals of the implicit surface.

Key words: Mesh evolution – Implicit surfaces

1 Introduction

Implicit surfaces are widely used in various applications, including engineering [26], computer graphics [6], and mathematics [28]. Given a field function $w = f(x, y, z)$, an implicit surface is defined as an isosurface (level set) of the field function $f(x, y, z) = \text{const}$.

For visualization purposes, polygonization of an implicit surface is often required. It includes sampling the function at selected points, estimating the positions of the mesh vertices, and connecting them to form polygons. This usually results in space aliasing (faceting) which is a common problem arising whenever some continuous object is represented by a set of discrete samples. Such faceting is most pervasive for shapes with sharp features (edges, corners, spikes, etc.).

In this paper, we develop an approach for accurate polygonization of implicit surfaces with sharp features. Given an implicit surface and its (usually rough) initial triangulation, we define a mesh evolution process initialized by the triangulation and evolving the mesh toward the implicit surface. The process also fits the mesh normals to implicit surface normals and therefore the evolving mesh approaches a high-quality polygonization of the implicit surface. Figure 1 shows an initial triangulation of an implicit surface with sharp edges (left) and a triangulated surface obtained after completing the mesh evolution process (right).

Sharp features, edges and corners, appear naturally if the implicit surface is constructed using set operations (union, intersection, difference). The defining function for such surface can be obtained by applying min/max functions or more general R -functions [23, 25, 26, 29] to the defining functions of the arguments of set operations. If the binary tree of operations is available, then the sharp features can be found by a numerical method that analyzes the functions of both arguments for each operation [36]. However, if the resulting function is evaluated by a procedure making the tree of operations not available to the application, this method is not applicable. An elegant approach for fast feature-sensitive surface extraction from volume data described by directed distance fields was recently proposed in [14]. Nevertheless we think that the traditional single function representation will remain to be a basis for the most of implicit modeling tools.

One possible antialiasing approach consists of adaptive-resolution polygonization, in which the sampling rate adapts to the changing feature size (curvature

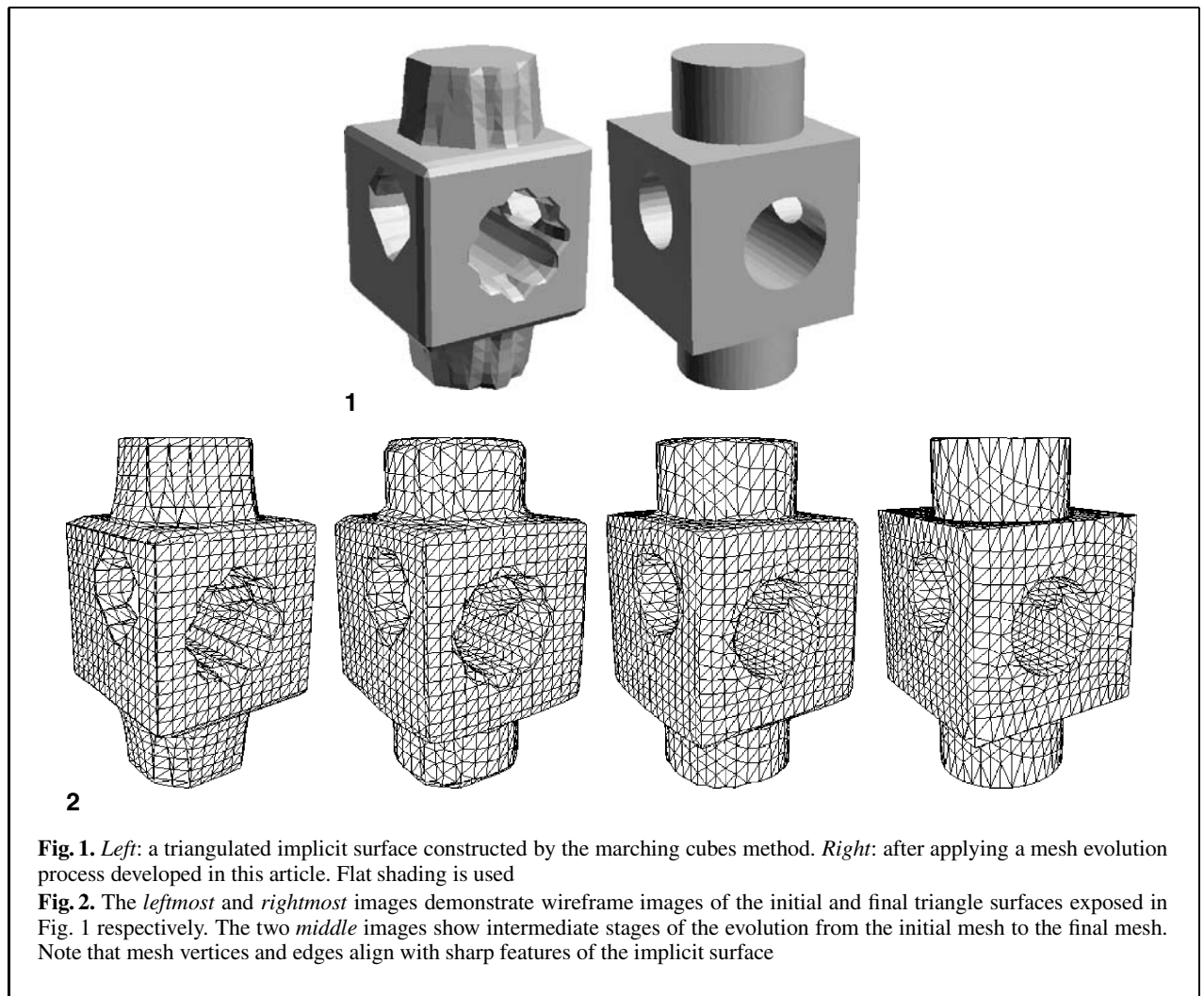


Fig. 1. *Left:* a triangulated implicit surface constructed by the marching cubes method. *Right:* after applying a mesh evolution process developed in this article. Flat shading is used

Fig. 2. The *leftmost* and *rightmost* images demonstrate wireframe images of the initial and final triangle surfaces exposed in Fig. 1 respectively. The two *middle* images show intermediate stages of the evolution from the initial mesh to the final mesh. Note that mesh vertices and edges align with sharp features of the implicit surface

scale) of a surface being polygonized [5, 11] (see also references therein).

Another remedy to reduce apparent faceting is smoothing. To date, the most powerful approaches for smoothing polygonal meshes are geometric signal processing [32, 33] and mesh evolution by diffusion flows [3, 8, 9, 16, 21, 28]. However, these smoothing techniques explore only local mesh properties and, therefore, are not best suited for isosurface smoothing when the original field function $w = f(x, y, z)$ is available.

Our basic idea is simple. Given a field function $w = f(x, y, z)$ and an initial polygonization (a triangle mesh) of the implicit surface $f(x, y, z) = 0$, we smooth the mesh with simultaneous minimization of the following

- the distances from the mesh vertices to the isosurface; and
- disparities between the isosurface normals and mesh normals.

In this work, we improve our previous dynamic mesh approach [22] developed for optimizing polygonizations of implicit surfaces with sharp features.

We act on the vertices of the evolving mesh with three forces. Two forces optimize positions of the vertices according to the values of the function and its gradient at the vertices. The third force improves mesh regularity. The combination of these three forces allows us to achieve an accurate approximation of the implicit surface by a high-quality mesh.

The leftmost and rightmost images of Fig. 2 demonstrate wireframe images of the initial and final trian-

gle surfaces exposed in Fig. 1 respectively. The two middle images of Fig. 2 show intermediate stages of the evolution from the initial mesh to the final mesh. Note that mesh vertices and edges align with sharp features of the implicit surface.

The idea to use meshes whose vertices act as a dynamic particle system was proposed in [10, 12] for 2D and in [17] for 3D and since then has been extensively used for segmentation and tracking in multidimensional images and volume data (see, for example, [4, 15, 18] for recent achievements in shape modeling and analysis with deformable contours and surfaces). Recently, dynamic meshes were applied to surface extraction from distance volume datasets [35]. However, in contrast to previous work, our method is developed to obtain an accurate mesh approximation of an implicit surface with sharp edges and corners. It is achieved by a simultaneous adjustment of the mesh vertices and mesh normals.

2 Mesh evolution toward implicit surface

Consider a family of meshes M^n evolving according to the following equation:

$$M^{n+1} = \mathcal{D}(M^n) = M^n + \mathbf{D}(M^n), \quad (1)$$

where $\mathcal{D}$ is a mesh-updating operator according to displacement vectors $\mathbf{D}$ defined at the mesh vertices. In many situations Eq. (1) can be considered an explicit approximation of an evolution of a family of smooth surfaces.

Consider an implicit surface $S = \{f(x, y, z) = 0\}$ and its polygonization M^0 . We want to construct a mesh-updating operator $\mathcal{D}$ such that the mesh evolution process of Eq. (1), with M^0 taken as the initial condition, converges to the surface, $M^n \rightarrow S$, as $n \rightarrow \infty$.

We implement the operator $\mathcal{D}$ satisfying the above requirement as the superposition of three mesh-updating operators (mesh flows) depending on f and local mesh properties:

$$\mathcal{D} = \mathcal{Z} \circ \mathcal{R} \circ \mathcal{N}.$$

These local operators are defined by the following vertex update procedures

$$\mathcal{N}\text{-flow} : P_{\text{new}} \leftarrow P_{\text{old}} + \mathbf{N}(P_{\text{old}}, \nu(P_{\text{old}}), f), \quad (2)$$

$$\mathcal{Z}\text{-flow} : P_{\text{new}} \leftarrow P_{\text{old}} + \mathbf{Z}(P_{\text{old}}, \nu(P_{\text{old}}), f), \quad (3)$$

$$\mathcal{R}\text{-flow} : P_{\text{new}} \leftarrow P_{\text{old}} + \mathbf{R}(P_{\text{old}}, \nu(P_{\text{old}})), \quad (4)$$

where $\nu(P)$ denotes the set of neighbors of vertex P in M , and $\mathbf{N}$, $\mathbf{Z}$, $\mathbf{R}$ are vector functions (forces).

In the subsequent subsections we define these vector functions such that $\mathcal{N}$ -flow corrects the mesh normals according to the implicit surface normals, $\mathcal{Z}$ -flow pushes mesh vertices toward the implicit surface $f(x, y, z) = 0$, and $\mathcal{R}$ -flow equalizes the mesh-sampling rate.

Figure 4 demonstrates how these mesh updating operators (flows) act on a triangle mesh. The top row demonstrates fragments of the triangulated block model shown in the left image of Fig. 1. The bottom row shows the same fragments after a number of repeated applications of the operators $\mathcal{N}$, $\mathcal{Z}$, and $\mathcal{R}$.

2.1 Moving mesh vertices toward the implicit surface

The simplest way to move mesh vertices toward a given implicit surface $f(x, y, z) = 0$ is to introduce an attracting force defined via the gradient of f . In our implementation we define $\mathbf{Z}$ -force at a mesh vertex P by

$$\begin{aligned} \mathbf{Z}(P, \nu(P, f)) &= -\tau A(P) \nabla [f(P)^2] \\ &= -2\tau A(P) f(P) \nabla f(P), \end{aligned}$$

where ∇ denotes the gradient, $A(P)$ is the sum of areas of all mesh triangles incident with P , and τ is a small positive parameter. Since the implicit surface $f(x, y, z) = 0$ is the minimal level set of the function $w = f(x, y, z)^2$, $\mathbf{Z}$ -force defined as above moves the mesh vertices toward the surface.

A similar force $-2\tau \text{sign}(f) \nabla f$ was used in [34] in connection with adaptive-resolution polygonization of implicit surfaces.

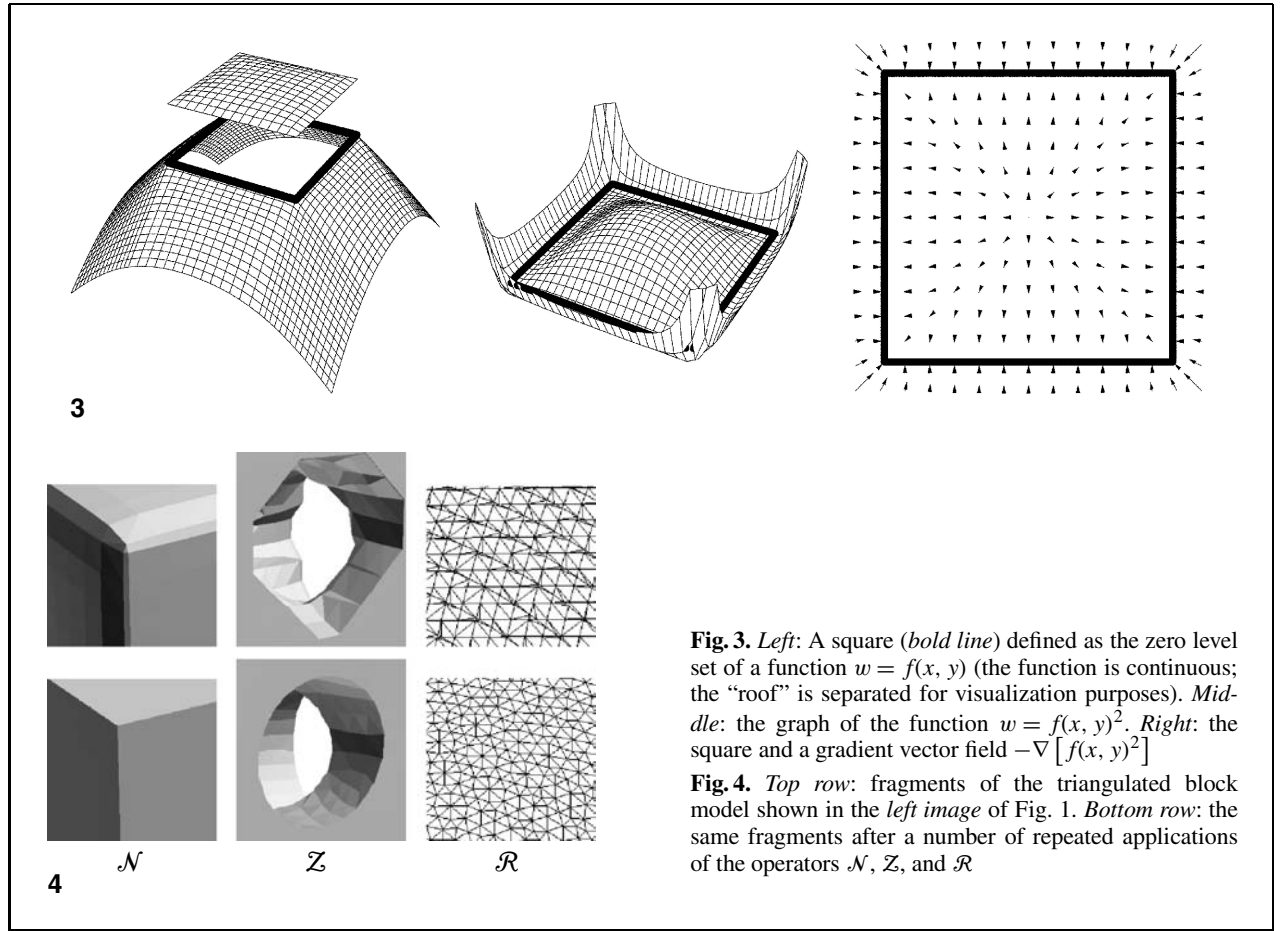
Figure 3 demonstrates a 2D level set $f(x, y) = 0$ defining a square and the associated $\mathbf{Z}$ -force.

The value of τ is chosen to ensure numerical stability of Eq. (3). Based on an analogy with a stability analysis for the first-order linear partial differential equations (the Courant–Friedrichs–Levy stability criterion [24]), we select τ such that

$$\max(A|f \nabla f|) \tau < c,$$

where c is a constant independent of the mesh and function f . In our current implementation, the time-step size τ is defined by

$$\tau = \frac{1}{100 \max(A|f \nabla f|)}. \quad (5)$$



The gradient ∇f can be computed analytically only for simple functions $w = f(x, y, z)$; so, we use standard central-difference formulae to estimate the gradient.

2.2 Mesh optimization according to implicit surface normals

A better approximation of the implicit surface $f(x, y, z) = 0$ is achieved if we modify positions of the mesh vertices such that the triangle normals are close to corresponding implicit-surface normals. Figure 5 illustrates advantages of such mesh optimization. The left image shows approximating a smooth curve by a polygonal line whose vertices are close to the curve. The right image demonstrates the same curve approximated by a polygonal line whose vertices and normals are close to curve points and normals.

An appropriate error function will be introduced and discussed in Sect. 2.4.

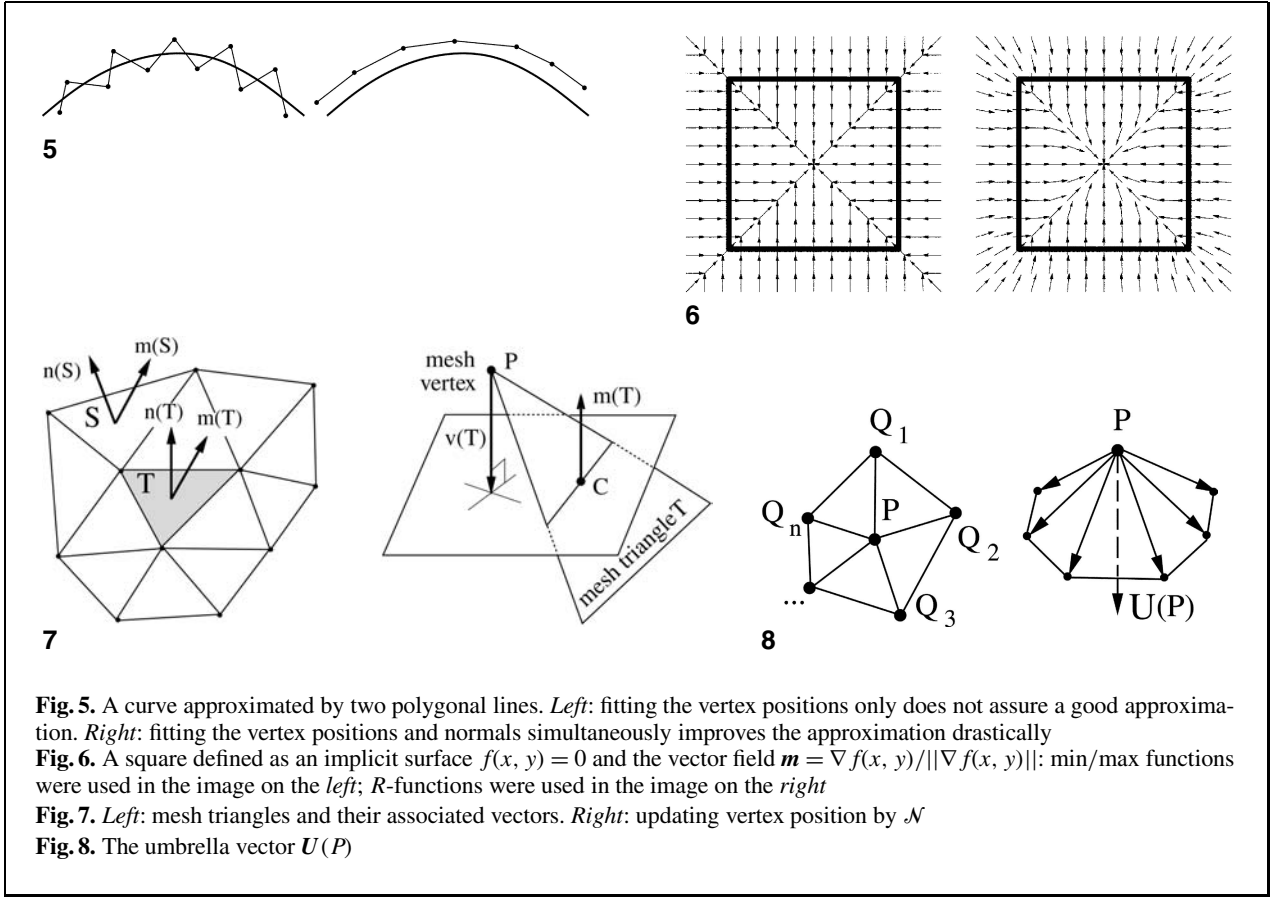
Consider a unit vector field

$$\mathbf{m}(x, y, z) = \frac{\nabla f(x, y, z)}{\|\nabla f(x, y, z)\|}.$$

It is orthogonal to the implicit surface $f(x, y, z) = 0$ at the implicit-surface points. Figure 6 presents a 2D example: a square defined as an implicit surface $f(x, y) = 0$ and its associated unit vector field $\mathbf{m}(x, y)$.

The figure demonstrates also that smoothness of the normalized gradient vector field $\mathbf{m}$ depends on functions chosen for basic Boolean operations. We use R -functions [25, 26, 30] since they have better differential properties than the commonly used min/max functions.

We want to move mesh vertices such that for every mesh triangle T the mesh normal $\mathbf{n}(T)$ becomes closer to $\mathbf{m}(T) = \nabla f(C)/\|\nabla f(C)\|$, where C is the



centroid of T . It is achieved by the $\mathcal{N}$ vertex update operator Eq. (2) with

$$\mathcal{N}(P, \mathbf{v}(P), f) = \frac{1}{\sum A(T)} \sum A(T) \mathbf{v}(T),$$

where $\mathbf{v}(T) = \left[\frac{\overrightarrow{PC} \cdot \mathbf{m}(T)}{\|\mathbf{m}(T)\|} \right] \mathbf{m}(T)$ is the projection of the vector $\overrightarrow{PC}$ on the $\mathbf{m}(T)$ direction, $A(T)$ denotes the area of T , and the summations are taken over all P -incident triangles (Fig. 7).

A similar vertex update operator was used in [21] for crease enhancement purposes.

2.3 Mesh relaxation

To improve mesh regularity, we use the tangential component of the Laplacian smoothing flow. The Laplacian flow, in its simplest form, moves repeatedly each mesh vertex by a displacement equal to

a positive scale factor times the average of the neighboring vertices. Consider a mesh vertex P and its neighbors $Q_1, \dots, Q_n$. The Laplacian flow is determined by the umbrella vector $\mathbf{U}$, as shown in Fig. 8, given by

$$\mathbf{U}(P) = \frac{1}{n} \sum_{i=1}^n Q_i - P.$$

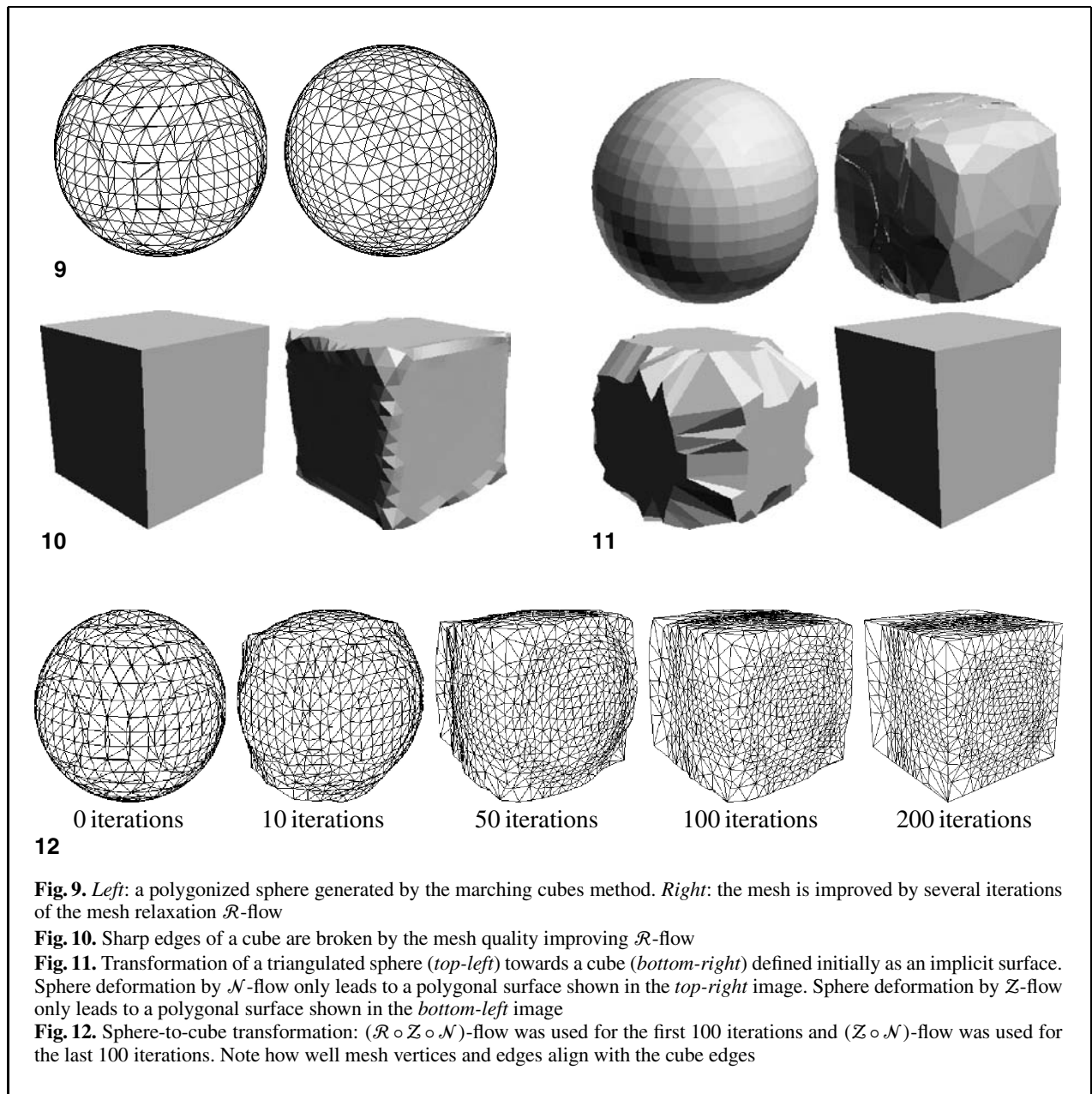
The vertex update operator $\mathcal{R}$ equalizing the mesh-sampling rate is defined by Eq. (4) with

$$\mathcal{R}(P, \mathbf{v}(P)) = C [\mathbf{U} - (\mathbf{U} \cdot \mathbf{n}) \mathbf{n}], \quad (6)$$

where $\mathbf{n}$ is the mesh normal at vertex P and C is a positive constant. In our current implementation, we use $C = 0.1$.

Figure 9 demonstrates how the tangential component of the umbrella operator improves the mesh-sampling quality.

However, the $\mathcal{R}$ -flow operator, Eq. (4), does not preserve sharp edges, as is demonstrated in Fig. 10.



Thus, when dealing with an implicit surface with sharp features, we have to switch off the mesh-quality-improving $\mathcal{R}$ -flow when the flow interferes with the mesh-to-surface convergence process.

Figure 11 demonstrates how the $\mathcal{N}$ -flow operator, Eq. (2), and the $\mathcal{Z}$ -flow operator, Eq. (3), transform a polygonized sphere (top left image) toward a cube (bottom-right image) defined as an implicit surface. If only $\mathcal{N}$ -flow is applied, the sphere transforms to

a polygonal surface shown in the top-right image. If the mesh is moved by $\mathcal{Z}$ -flow only, the sphere transforms to a polygonal surface shown in the bottom-left image. A perfect metamorphosis of the sphere to the cube is achieved if at first we use the composition of the three introduced flows $\mathcal{R} \circ \mathcal{Z} \circ \mathcal{N}$ and then shift to the flow $\mathcal{Z} \circ \mathcal{N}$ according to an analysis of an error function defined and studied in the next section (Fig. 12).

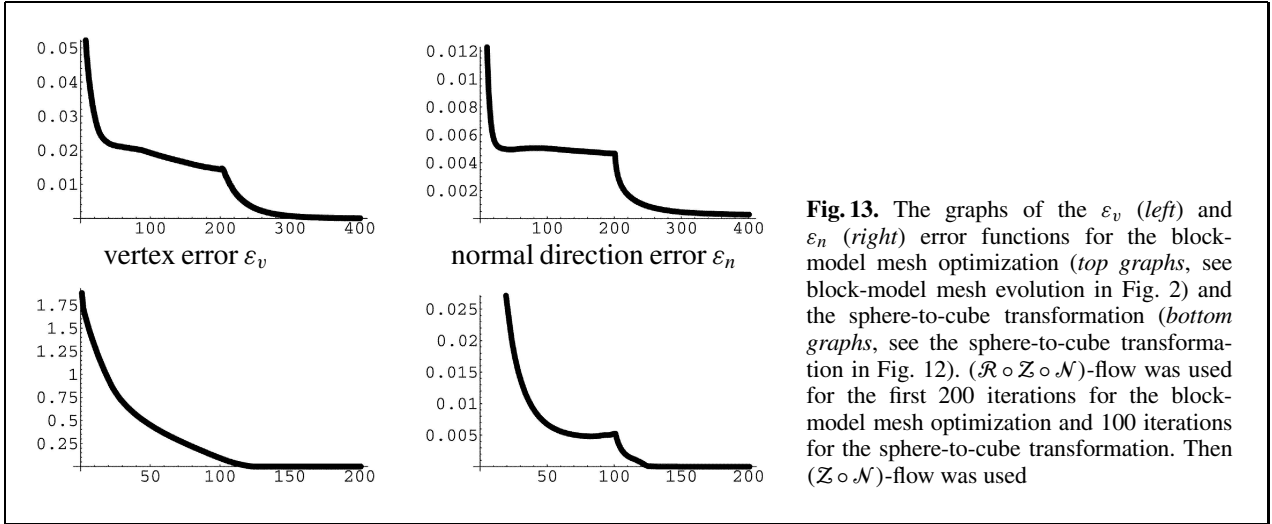


Fig. 13. The graphs of the ε_v (left) and ε_n (right) error functions for the block-model mesh optimization (*top graphs*, see block-model mesh evolution in Fig. 2) and the sphere-to-cube transformation (*bottom graphs*, see the sphere-to-cube transformation in Fig. 12). $(\mathcal{R} \circ \mathcal{Z} \circ \mathcal{N})$ -flow was used for the first 200 iterations for the block-model mesh optimization and 100 iterations for the sphere-to-cube transformation. Then $(\mathcal{Z} \circ \mathcal{N})$ -flow was used

2.4 Error analysis

To analyze how close the evolving mesh M^n approaches the implicit surface $f(x, y, z) = 0$, as $n \rightarrow \infty$, we introduce error estimator functions ε_v and ε_n , where ε_v measures the deviation of the mesh vertices from $f(x, y, z) = 0$ and ε_n characterizes the deviation of the mesh triangle normals from the normalized gradient vector field $\nabla f / \|\nabla f\|$.

The squared distance from a mesh vertex P to the implicit surface $f(x, y, z) = 0$ can be estimated by the first-order approximation $f(P)^2 / \|\nabla f(P)\|^2$, as in [31]. So we define ε_v by

$$\varepsilon_v = \frac{1}{3 \sum_{\text{all } T} A(T)} \sum_{\text{all } P} \left(\frac{f(P)^2}{\|\nabla f(P)\|^2} \sum_{T \in \text{inc}(P)} A(T) \right),$$

where the sum is taken over $\text{inc}(P)$, all mesh triangles T incident with P , and $A(T)$ denotes the area of T .

The deviation of the mesh triangle normals from $\nabla f / \|\nabla f\|$ is estimated by

$$\varepsilon_n = \frac{1}{\sum_{\text{all } T} A(T)} \sum_{\text{all } T} [A(T) (1 - \mathbf{n}(T) \cdot \mathbf{m}(T))],$$

where the summations are taken over all the mesh triangles T , $\mathbf{n}(T)$ is the unit normal of T , $\mathbf{m}(T) = \nabla f(C) / \|\nabla f(C)\|$, where C is the centroid of T .

Note that ε_v mimics the sum of squared distances from the mesh vertices to the implicit surface, and

ε_n gives a weighted sum (integral) of squared differences between the mesh and surface normals since

$$(\mathbf{n} - \mathbf{m})^2 = 2(1 - \mathbf{n} \cdot \mathbf{m}).$$

Note that $\varepsilon_v + \varepsilon_f$ mimics the squared H^1 norm used widely in the theory of partial differential equations [1] and for error analysis of finite-element methods [7].

As we noted in the previous section, the mesh motion by the tangential component of the umbrella operator ($\mathcal{R}$ -flow) destroys sharp edges (Fig. 10). We switch off the mesh regularization by $\mathcal{R}$ -flow after ε_n has stabilized.

Figure 13 demonstrates the behaviors of ε_v and ε_n for the block-model mesh evolution (Figs. 1 and 2) and the sphere-to-cube transformation (Fig. 12).

3 Further improvements

3.1 Modified $\mathcal{Z}$ - and $\mathcal{N}$ -flows

Alternatively, at the final stage of our mesh optimization procedure, a combination of modified $\mathcal{Z}$ - and $\mathcal{N}$ -flows can be used instead of $(\mathcal{Z} \circ \mathcal{N})$ -flow.

Consider a mesh vertex P , a mesh triangle T incident with P , and two other vertices of the triangle, P_1 and P_2 . Let $C = (P + P_1 + P_2)/3$ denote the triangle centroid. Let us measure the deviation of the triangle normal $\mathbf{n}(T)$ from $\mathbf{m}(C)$ by

$$e_n(T) = A(T)(\mathbf{n} - \mathbf{m})^2 = 2A(T)(1 - \mathbf{n} \cdot \mathbf{m}) \quad (7)$$

and define a local energy associated with P by

$$e_n(P) = \sum_{T \in \text{inc}(P)} e_n(T),$$

where the sum is taken over all mesh triangles T incident with P .

The $\tilde{\mathcal{N}}$ -flow updates the position of P such that the local energy $e_n(P)$ takes its minimal value. We minimize $e_n(P)$ by a gradient descent method. Let $\tilde{\mathcal{N}} = \tilde{\mathcal{N}}(P, \nu(P), f)$ be the antigradient $-\nabla$ of the energy $e_n(P)$ with respect to $P(x, y, z)$. The updated position of P is approximated by five consecutive iterations

$$P_{\text{new}} \leftarrow P_{\text{old}} + \frac{1}{5 \deg(P)} \tilde{\mathcal{N}}(P_{\text{old}}, \nu(P_{\text{old}}), f)$$

started from the initial position of P . Here $\deg(P)$ is the degree of vertex P .

Computing the antigradient of $e_n(P)$ with respect to $P(x, y, z)$ is trivial if we know the antigradients of $e_n(T)$ for the triangles T surrounding P . Computing the antigradient of $e_n(T)$ is simple. The antigradient of the area of the triangle PP_1P_2 is given by $e|P_1P_2|/2$, where e is the unit vector defining the height direction from P to the opposite side P_1P_2 , as seen in Fig. 14. Now let us note that $e_n(T)$ is proportional to the difference between the area of PP_1P_2 and the area of the projection of PP_1P_2 onto a plane orthogonal to $m(C)$. Thus the antigradient of $e_n(T)$ has clear geometric meaning and can be easily found. The $\tilde{\mathcal{Z}}$ -flow is defined by $\tilde{\mathcal{Z}}$ -force given by

$$\tilde{\mathcal{Z}}(P, f) = -2\tau \sum_{T \in \text{inc}(P)} A(T) \nabla f(C) f(C)/3,$$

where the step-size parameter τ is given by Eq. (5). It can be considered as a simplified antigradient of the energy

$$e_v(T) = \sum_{T \in \text{inc}(P)} e_v(T), \quad \text{where}$$

$$e_v(T) = A(T) f(C)^2.$$

Figure 15 compares $\mathcal{R} \circ \mathcal{Z} \circ \mathcal{N} + \mathcal{Z} \circ \mathcal{N}$ and $\mathcal{R} \circ \mathcal{Z} \circ \mathcal{N} + \tilde{\mathcal{Z}} \circ \tilde{\mathcal{N}}$ mesh optimization procedures. (The computational times were measured on a 500 MHz Mobile Pentium III computer.) The latter procedure requires less iterations, works faster, and produces better reconstruction of sharp edges. The visual

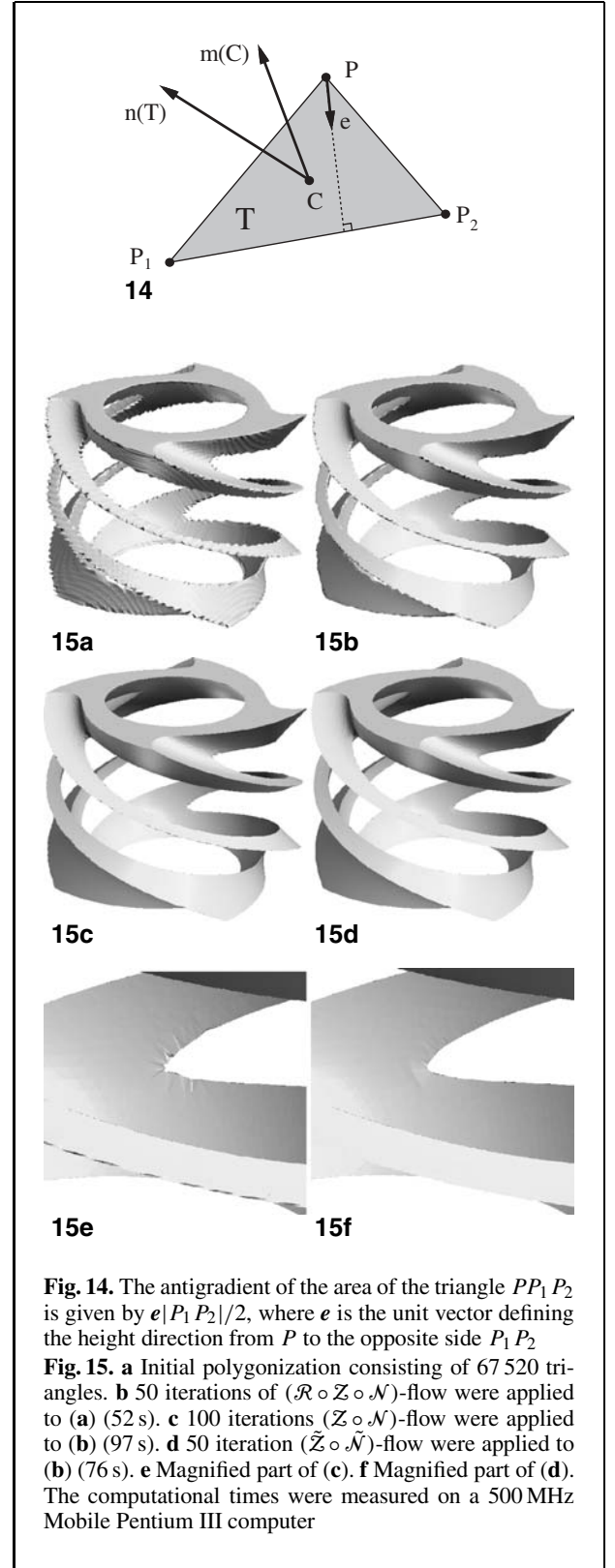


Fig. 14. The antigradient of the area of the triangle PP_1P_2 is given by $e|P_1P_2|/2$, where e is the unit vector defining the height direction from P to the opposite side P_1P_2

Fig. 15. **a** Initial polygonization consisting of 67 520 triangles. **b** 50 iterations of $(\mathcal{R} \circ \mathcal{Z} \circ \mathcal{N})$ -flow were applied to **(a)** (52 s). **c** 100 iterations $(\mathcal{Z} \circ \mathcal{N})$ -flow were applied to **(b)** (97 s). **d** 50 iteration $(\tilde{\mathcal{Z}} \circ \tilde{\mathcal{N}})$ -flow were applied to **(b)** (76 s). **e** Magnified part of **(c)**. **f** Magnified part of **(d)**. The computational times were measured on a 500 MHz Mobile Pentium III computer

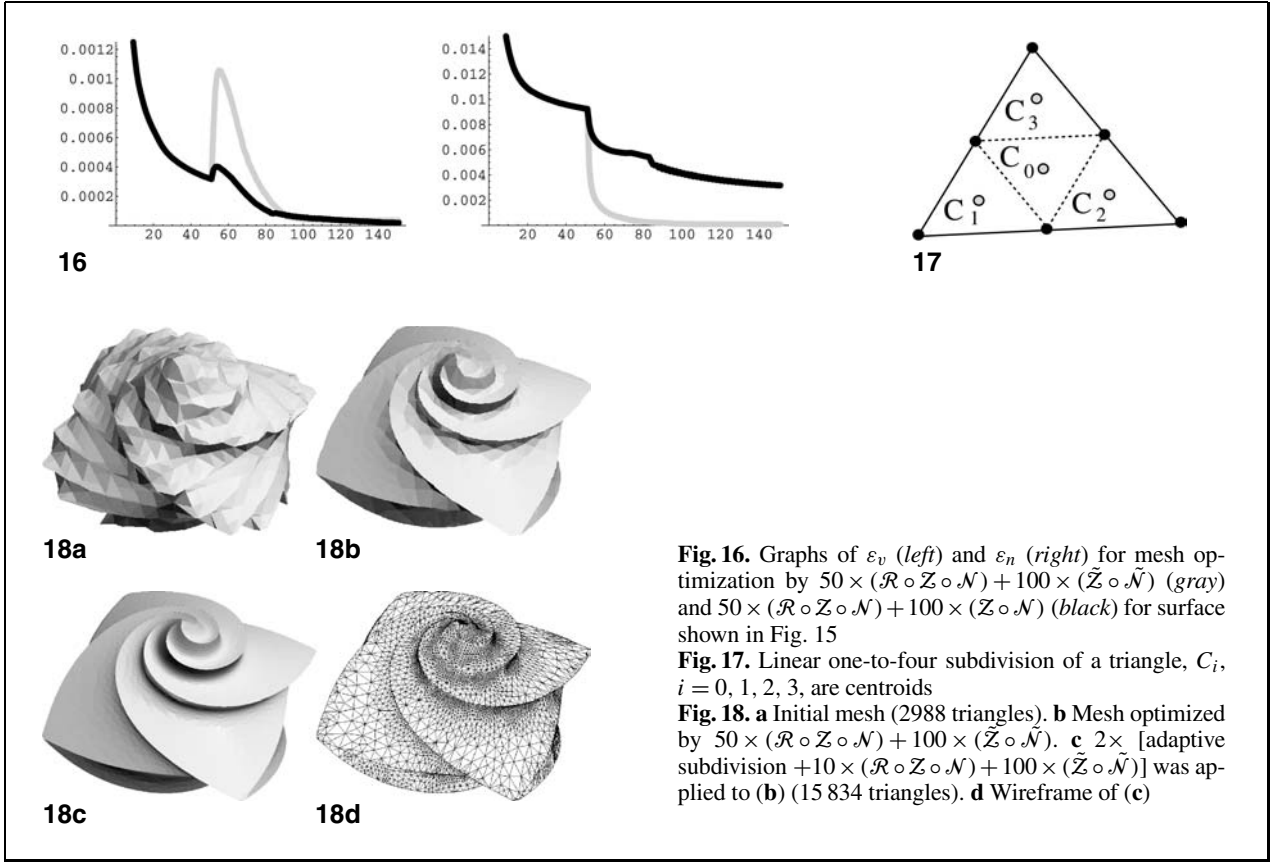


Fig. 16. Graphs of ε_v (left) and ε_n (right) for mesh optimization by $50 \times (\mathcal{R} \circ \mathcal{Z} \circ \mathcal{N}) + 100 \times (\tilde{\mathcal{Z}} \circ \tilde{\mathcal{N}})$ (gray) and $50 \times (\mathcal{R} \circ \mathcal{Z} \circ \mathcal{N}) + 100 \times (\mathcal{Z} \circ \mathcal{N})$ (black) for surface shown in Fig. 15

Fig. 17. Linear one-to-four subdivision of a triangle, C_i , $i = 0, 1, 2, 3$, are centroids

Fig. 18. **a** Initial mesh (2988 triangles). **b** Mesh optimized by $50 \times (\mathcal{R} \circ \mathcal{Z} \circ \mathcal{N}) + 100 \times (\tilde{\mathcal{Z}} \circ \tilde{\mathcal{N}})$. **c** $2 \times$ [adaptive subdivision + $10 \times (\mathcal{R} \circ \mathcal{Z} \circ \mathcal{N}) + 100 \times (\tilde{\mathcal{Z}} \circ \tilde{\mathcal{N}})$] was applied to **(b)** (15 834 triangles). **d** Wireframe of **(c)**

comparison is also confirmed by the comparison of graphs of the error functions ε_v and ε_n , as seen in Fig. 16.

3.2 Adaptive mesh subdivision

If the initial polygonization is not dense enough, small surface features cannot be well reconstructed. So we use linear one-to-four subdivision of those mesh triangles T where the mesh normals $\mathbf{n}(T)$ have large deviations from the implicit-surface normals $\mathbf{m} = \nabla f / \|\nabla f\|$.

Consider a triangle T and its imaginary one-to-four subdivision into four triangles T_i with centroids C_i , $i = 0, 1, 2, 3$, as seen in Fig. 17.

It is convenient to measure the deviation of $\mathbf{n}(T)$ from the implicit surface normals by a slight modification of Eq. (7)

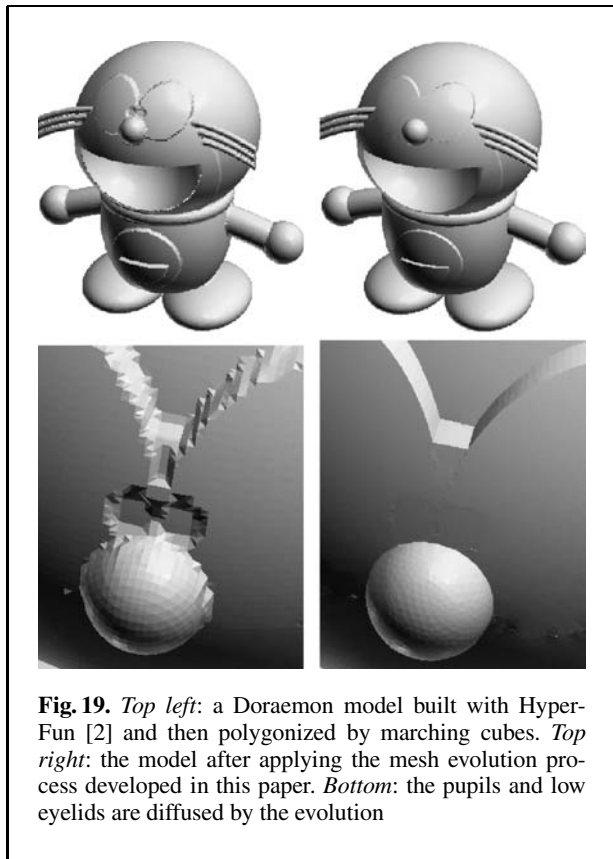
$$\hat{\varepsilon}_n(T) = A(T) \sum_{i=1}^4 (1 - |\mathbf{n}(T) \cdot \mathbf{m}(C_i)|),$$

where $A(T)$ is the area of triangle T . If $\hat{\varepsilon}_n(T)$ is greater than a user-specified threshold, triangle T is subdivided.

The adaptive subdivision procedure allows us to generate high-quality polygonal approximations of complex implicit surfaces starting from a coarse initial mesh, as seen in Fig. 18.

4 Conclusion and future work

We have presented a novel approach for accurate polygonization of implicit surfaces with sharp features. The approach is based on mesh evolution toward a given implicit surface with simultaneous control of the mesh vertex positions and mesh normals. In this paper, we have not considered the case when a voxelized data set is given instead of a field function $w = f(x, y, z)$ defined for every (x, y, z) . We refer interested readers to our recent work [19], where optimizing isosurfaces with boundaries was also studied.



The presented implementation of the approach can be improved in many directions. For example, if the sampling rate of the initial mesh is comparable with the size of a feature of a given implicit surface, then the feature can be lost during the mesh-to-surface evolution process. Figure 19 demonstrates a Doraemon¹ model built with HyperFun [2] and then polygonized by the marching cubes method (top left image) and the result after applying the mesh-to-surface evolution process developed in this paper (top right image). Note that the mesh evolution removes the cat's pupils (see the bottom images). One way around this problem consists in developing a better mesh subdivision procedure. A combination with a dynamic mesh connectivity approach [13] may be also useful.

The speed of convergence of a dynamic mesh toward an implicit surface $f(x, y, z) = 0$ with sharp features depends on differential properties of the function

$w = f(x, y, z)$. The min/max functions commonly used [27] for the basic Boolean operations, union and intersection, have poor differential properties. For example, $\max(x, y)$ and $\min(x, y)$ are not differentiable along the line $x = y$. Whereas certain R -functions [25, 26, 30] also define the union and intersection and, in addition, possess much better differential properties. Figure 6 demonstrates advantages of R -functions over the min/max functions. R -functions are implemented in the HyperFun geometric modeling language and supporting software [2]. A variational approach to optimizing polygonized implicit surfaces constitutes another promising direction for future research. Our first steps in this direction [20] are very encouraging.

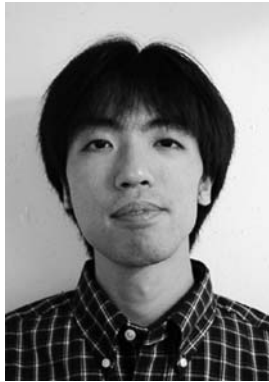
References

1. Adams RA (1975) Sobolev spaces. Academic Press, New York
2. Adzhiev V, Cartwright R, Fausett E, Ossipov A, Pasko A, Savchenko V (1999). HyperFun project: a framework for collaborative multidimensional F-rep modeling. In: Hughes J, Schlick C (eds) Implicit Surfaces '99, Eurographics/ACM SIGGRAPH Workshop, September 1999. ACM SIGGRAPH, New York, pp 59–69; Visit also <http://www.hyperfun.org>
3. Bajaj C, Xu G (2001) Anisotropic diffusion of noisy surfaces and noisy functions on surfaces. In: Proceedings of Information Visualization 2001 – Computer Aided Geometric Design. IEEE Computer Society Press, Los Alamitos, Calif.
4. Blake A, Isard M (1998) Active contours. Springer, Berlin Heidelberg New York
5. Bloomenthal J (1997) Surface tiling. In: Bloomenthal J (ed) Introduction to implicit surfaces. Morgan Kaufmann, San Francisco, Calif.
6. Bloomenthal J (ed) (1997) Introduction to implicit surfaces. Morgan Kaufmann, San Francisco, Calif.
7. Brenner SC, Scott LR (1994) The mathematical theory of finite element methods. Springer, Berlin Heidelberg New York
8. Clarenz U, Diewald U, Rumpf M (2000) Anisotropic geometric diffusion in surface processing. In: Proceedings of IEEE Visualization 2000. IEEE Computer Society Press, Los Alamitos, Calif., pp 397–405
9. Desbrun M, Meyer M, Schröder P, Barr AH (1999) Implicit fairing of irregular meshes using diffusion and curvature flow. In: Proceedings of SIGGRAPH '99. ACM SIGGRAPH, New York, pp 317–324
10. Durbin R, Willshaw D (1987) An analogue approach to the traveling salesman problem using an elastic net method. Nature 326:689–691
11. Karkanis T, Stewart AJ (2001) Curvature-dependent triangulation of implicit surfaces. IEEE Comput Graph Appl 21(2):60–69

¹ Doraemon is a round cat-style robot, one of the most famous Japanese manga characters.

12. Kass M, Witkin A, Terzopoulos D (1987) Snakes: active contour models. *Int J Comput Vis* 1(4):321–331
13. Kobbelt L, Bareuther T, Seidel HP (2000) Multi-resolution shape deformations for meshes with dynamic vertex connectivity. *Computer Graphics Forum (Eurographics 2000 issue)* 19(3):249–260
14. Kobbelt LP, Botsch M, Schwanecke U, Seidel H-P (2001) Feature sensitive surface extraction from volume data. In: *Proceedings of SIGGRAPH 2001*. ACM SIGGRAPH, New York, pp 57–66
15. McInerney T, Terzopoulos D (1999) Topology adaptive deformable surfaces for medical image volume segmentation. *IEEE Trans Med Imag* 18(10):840–850
16. Meyer M, Desbrun M, Schröder P, Barr AH (2002) Discrete differential-geometry operators for triangulated 2-manifolds. In: *International Workshop on Visualization and Mathematics*, Berlin
<http://www.multires.caltech.edu/pubs/pubs.htm>
17. Miller JV, Breen DE, Lorensen WE, O'Bara RM, Wozny MJ (1991) Geometrically deformed models: a method for extracting closed geometric models from volume data. *Comput Graph (Proceedings of SIGGRAPH '91)* 25(4):217–226
18. Montagnat J, Delingette H, Scapel N, Ayache N (2000) Representation, shape, topology, and evolution of deformable surfaces: application to 3D medical image segmentation. Research report RR-3954. INRIA Sophia Antipolis, France
19. Ohtake Y, Belyaev AG (2001) Mesh optimization for polygonized isosurfaces. *Comput Graph Forum (Eurographics 2001 issue)* 20(3):368–376
20. Ohtake Y, Belyaev AG (2002) Dual/primal mesh optimization for polygonized implicit surfaces. In: *Proceedings of 7th ACM Symposium on Solid Modeling*, Saarbrücken, Germany. ACM Press, New York, pp 171–178
21. Ohtake Y, Belyaev AG, Bogaevski IA (2001) Mesh regularization and adaptive smoothing. *Comput-Aided Des* 33(11):789–800
22. Ohtake Y, Belyaev AG, Pasko AA (2001) Dynamic meshes for accurate polygonization of implicit surfaces with sharp features. In: *International Conference on Shape Modeling and Applications 2001*, Genova, Italy. IEEE Computer Society Press, Los Alamitos, Calif. pp 74–81
23. Pasko A, Adzhiev V, Sourin A, Savchenko V (1995) Function representation in geometric modeling: concepts, implementation and applications. *Vis Comput* 11(8):429–446
24. Press WH, Teukolsky SA, Vetterling WT, Flannery BP (1993) *Numerical recipes in C: the art of scientific computing*. Cambridge University Press, Cambridge, U.K.
25. Rvachev VL (1973) *The methods of the algebra of logic in mathematical physics (in Russian)*. Naukova Dumka, Kiev
26. Rvachev VL (1982) *Theory of R-functions and some of its applications (in Russian)*. Naukova Dumka, Kiev
27. Schroeder W, Martin K, Lorensen W (1998) *The visualization toolkit: an object-oriented approach to 3D graphics*. Prentice Hall, Upper Saddle Creek, N.J.
28. Sethian JA (1999) *Level set methods and fast marching methods*. Cambridge University Press, Cambridge, U.K.
29. Shapiro V (1994) Real functions for representation of rigid solids. *Comput Aided Geom Des* 11(2):152–175
30. Shapiro V, Tsukanov I (1999) Implicit functions with guaranteed differential properties. In: *Proceedings of the Fifth Symposium on Solid Modeling (SOLID MODELING '99)*, Ann Arbor, Michigan, ACM Press, New York, pp 852–857
31. Taubin G (1993) An improved algorithm for algebraic curve and surface fitting. In: *IEEE Proceedings of the Third International Conference on Computer Vision*. IEEE Computer Society Press, Los Alamitos, Calif., pp 658–665
32. Taubin G (1995) A signal processing approach to fair surface design. In: *Proceedings of SIGGRAPH '95*. ACM SIGGRAPH, New York, pp 351–358
33. Taubin G (2000) Geometric signal processing on polygonal meshes. In: *EUROGRAPHICS 2000: State of the Art Reports*, The Eurographics Association, pp 107–117
34. Velho L (1996) Simple and efficient polygonization of implicit surfaces. *J Graph Tools* 1(2):5–24
35. Wood ZJ, Desbrun M, Schröder P, Breen D (2000) Semi-regular mesh extraction from volumes. In: *Proceedings of IEEE Visualization 2000*. IEEE Computer Society Press, Los Alamitos, Calif., pp 275–282
36. Wyvill B and K. van Overveld. Polygonization of implicit surfaces with constructive solid geometry. *Int J Shape Modelling* 2(4):257–274

Photographs of the authors and their biographies are given on the next page.



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